

PI Current Controller (CC-ICL) — Differential-Algebraic Equations

Matas-Díaz et al., "A Systematic Small-signal ...", JMPSC 2025, §II.C.2, eqs. (17)–(20)

1. State variables $\mathbf{x}$ (4)

i_{sd}^*	— d-axis filtered current reference [LPF output] (eq. 18)
i_{sq}^*	— q-axis filtered current reference [LPF output] (eq. 18)
ξ_{id}	— Integral of d-axis current error (eq. 19b)
ξ_{iq}	— Integral of q-axis current error (eq. 19b)

2. Algebraic variables $\mathbf{y}$ (2)

v_{td}	— VSC terminal voltage, d-axis (eq. 19a)
v_{tq}	— VSC terminal voltage, q-axis (eq. 19a)

3. Parameters (default values, Table I)

k_{pi}^{cc}	1.25 [V/A]	Proportional gain of PI current controller
k_{ii}^{cc}	40.0 [V/(A·s)]	Integral gain of PI current controller
G_v	0.0 [S]	Virtual conductance
B_v	1.25 [S]	Virtual susceptance
τ_{lpf}	1.6 ms	LPF time constant
L_f	caller-supplied [H]	Total filter inductance

4. Differential equations $\mathbf{dx/dt} = \mathbf{f}(\mathbf{x}, \mathbf{u})$

Virtual admittance output (eq. 17, $\mathbf{e}^T = [0, E]^T$, $\mathbf{Y}_v = [[G_v, -B_v], [B_v, G_v]]$):

$$\begin{aligned} i_{s,d}^{va} &= -G_v v_{sd} - B_v (E - v_{sq}) \\ i_{s,q}^{va} &= -B_v v_{sd} + G_v (E - v_{sq}) \end{aligned}$$

LPF dynamics (eq. 18):

$$\begin{aligned} \dot{i}_{sd}^* &= \frac{i_{s,d}^{va} - i_{sd}^*}{\tau_{lpf}} \\ \dot{i}_{sq}^* &= \frac{i_{s,q}^{va} - i_{sq}^*}{\tau_{lpf}} \end{aligned}$$

PI integrators (eq. 19b):

$$\begin{aligned} \dot{\xi}_{id} &= i_{sd}^* - i_{sd} \\ \dot{\xi}_{iq} &= i_{sq}^* - i_{sq} \end{aligned}$$

5. Algebraic equations $\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u})$ (eq. 19a)

$$\begin{aligned} 0 &= v_{td} - [k_{pi}^{cc}(i_{sd}^* - i_{sd}) + k_{ii}^{cc} \xi_{id} + \omega_{vsg} L_f i_{sq} + v_{sd}] \\ 0 &= v_{tq} - [k_{pi}^{cc}(i_{sq}^* - i_{sq}) + k_{ii}^{cc} \xi_{iq} - \omega_{vsg} L_f i_{sd} + v_{sq}] \end{aligned}$$

6. Inputs $\mathbf{u}$ (provided by other subsystems)

v_{sd}, v_{sq}	— Grid-side inductor voltage, dq components
i_{sd}, i_{sq}	— Measured grid-side current, dq components
E	— Virtual EMF amplitude (from OCL)
ω_{vsg}	— VSG angular speed (from OCL)